

Time Sensitive Networking(TSN) Solution

Qbit Labs' TSN Solution delivers deterministic, low-latency Ethernet for industrial, automotive, and aerospace systems. Built for FPGA/ASIC, it ensures full IEEE TSN compliance with precise timing, traffic shaping, redundancy, and optional MACsec security—ideal for real-time, safety-critical applications like robotics, autonomous vehicles, and mission-critical control networks.

Features & Benefits

🔧 **Switching:** Wire-speed (10 Mbps–10 Gbps), up to 24 ports, 4K MAC/VLAN tables, Switching Capacity 30 GB per second, static and dynamic mac table

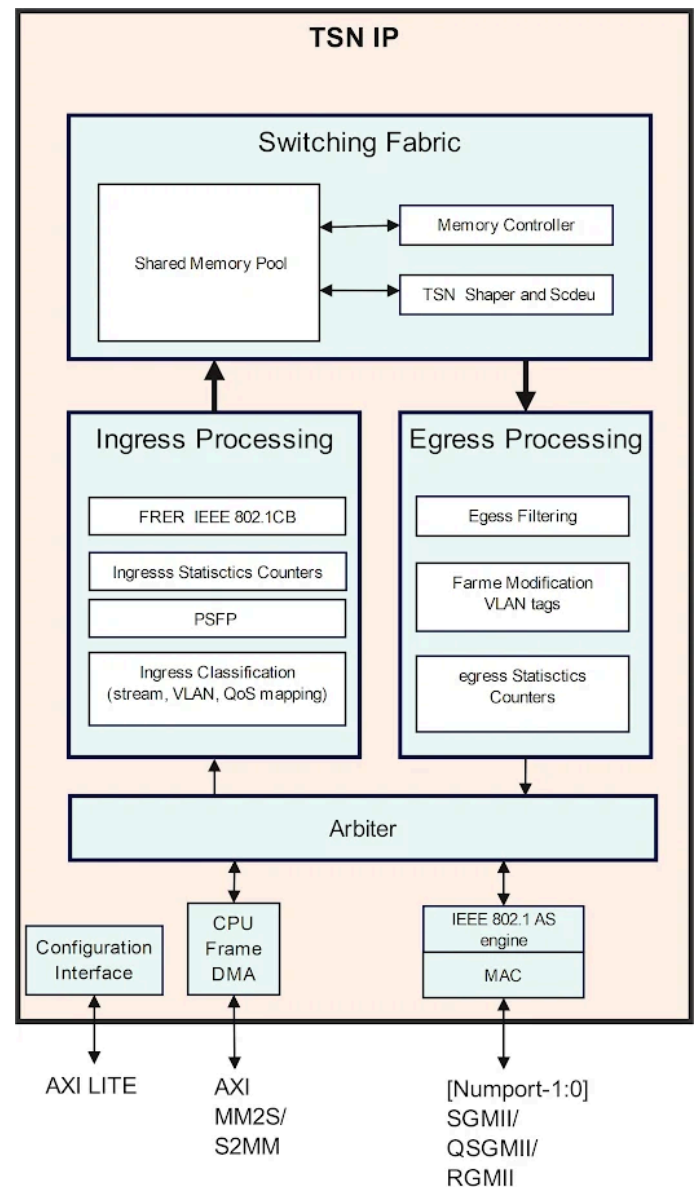
🕒 **TSN:** Time-aware shaping, frame preemption, PSFP, FRER

🔄 **Synchronisation:** gPTP offload, sub- μ s precision, 1PPS Input/Output

⚡ **QoS:** 8 priority classes, broadcast policer, priority based flow control, port rate limiting, egress/ingress filtering

🔧 Compliant Standards -

- IEEE 802.1Qbv
- IEEE 802.1Qbu/802.3br
- IEEE 802.1Qci
- IEEE 802.1Qcc
- IEEE 802.1Q
- IEEE 802.1ab
- IEEE 802.1AS-2020
- IEEE 802.1X
- IEEE 802.1Qbb
- IEEE802.1p
- IEEE 802.1dp
- 802.1DG



Target Applications

- 🏭 Industrial Automation & Control
- 🚗 Automotive Networking & ADAS
- ✈ Aerospace & Defense (safety-critical)
- 🎵 Audio Video Bridging (low-latency media)

Why Choose Us?

- 🚀 Rapid integration, faster time-to-market
- 📄 Clean, verified, documented RTL
- 👨‍💻 Direct access to experts

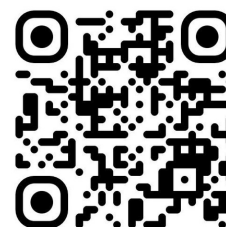


Technical Specifications

Category	Specification
Feature & Port Configurations	• Max Data Ports: 24 (configurable) • Max 10/100 Mbps Ports: 24 (configurable) • Max 1G Ports: 24 (configurable) • Max 10G Ports: 4 (configurable) • CPU Port: 1
L2 Switching	• Packet Buffer: 8Mb (configurable) • Static MAC Table: up to 256 entries • MAC Learning Table: up to 4096 entries • VLAN Table: up to 4096 entries • Layer 2 Multicast Port Mask: up to 1024
QoS & Security	• Egress Queues per Port: 8 • Ingress Queues per Port: 1 per priority • Static MAC Binding per Port: 16 • Learned MAC Binding per Port: 128 • Max Ingress ACL Rules: 2048 • Max Egress ACL Rules: 512
TSN	• PSFP Streams: 256 • PSFP Stream Filters: 256 • PSFP Stream Gates: 4 • TAS Gate Control Lists: 64 • TAS Entries per List: 256 • FRER Streams: 256
CPU Interface	• AXI LITE (register config): 1 • AXI MM2S (CPU Tx packets): 1 • AXI S2MM (CPU Rx packets): 1
Driver & Management	Driver: Linux/Windows
Management	HTTPS Web GUI, SNMPv3, NETCONF, SSH

Get in touch with us

Delivering Advanced IP Solutions
for Next-Gen Networking



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